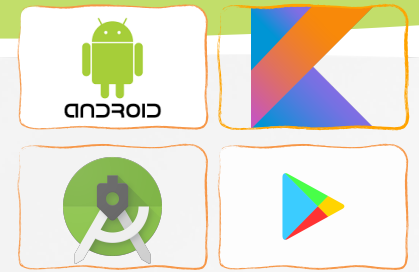




Software Development for Mobile Devices

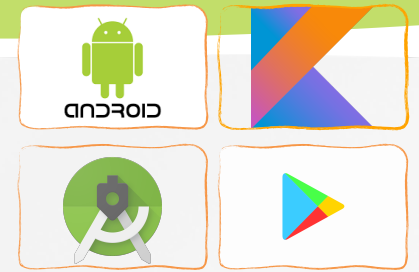
**(Based on Cis470 of
Cleveland State University)**

Nguyen Le Hoang Dung
nlhdung@fit.hcmus.edu.vn



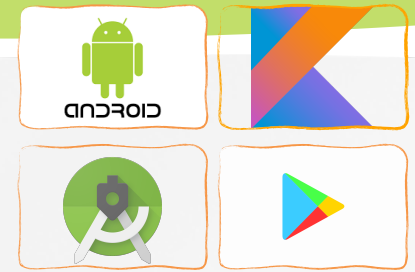
Content

- » Objectives
- » Lecturers
- » Development Environment
- » Grades
- » Policy
- » Quiz



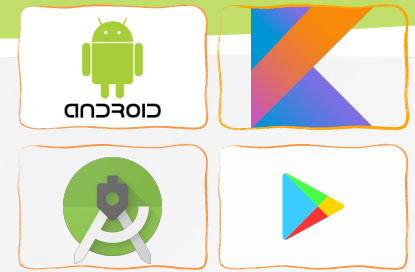
Objectives

- » In this class we will learn how to develop applications for the Android mobile platform
- » This class will be taught primarily using the Kotlin and/or Java languages.
- » <https://developer.android.com>



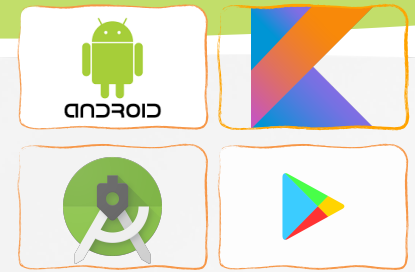
Objectives

- » Understand how design patterns such as model-view-controller or MVVM are used in software frameworks
- » Implement mobile software applications that use text and touch input as well as using input from sensors
- » Understand how to store data in a mobile application
- » Understand how to optimize performance of a mobile device application and use threading
- » Understand how to create background and foreground services
- » Understand how to send data to a remote server
- » Understand how to use maps, video, audio....



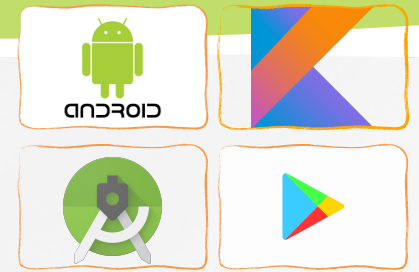
Main Content

- » Tools and development environments
- » Kotlin
- » How to design a Android app
- » App lifecycle
- » Views and layouts
- » Sensors
- » Persistence and databases
- » Creating background and foreground services



Main Content

- » Sending data to a server
- » Video and Audio
- » Maps and Location
- » Contacts and Calendars
- » Notification
- » Revenue and IAP
- » Multi-language
- » Advanced topics



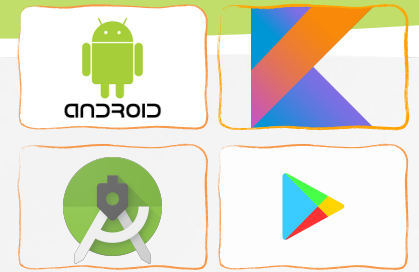
Lecturers

» Instructor:

» MSc. Nguyễn Lê Hoàng Dũng
(nlhdung@fit.hcmus.edu.vn)

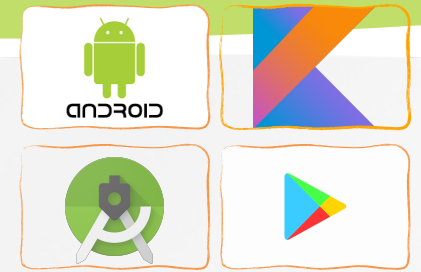
» TA:

» MSc. Trần Duy Quang



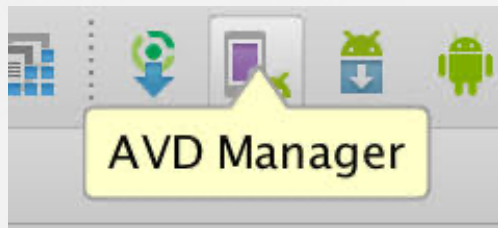
Lecturers (cont.)

- » Discussion Boards:
 - » Moodle
 - » Facebook Group
 - » Course resources



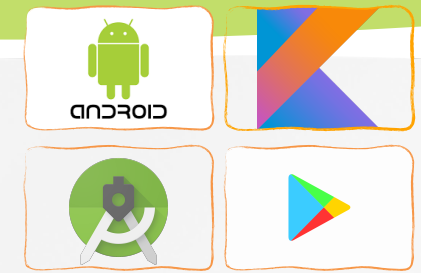
Development Environment

- » Platform: Android
- » Language: Kotlin
- » IDE and Emulators



- » Version control





Development Environment (cont.)

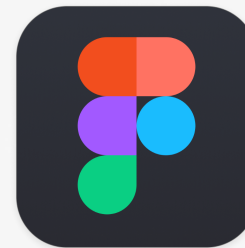
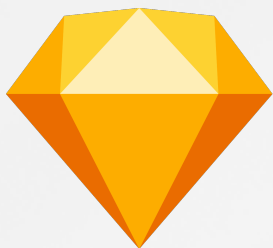
» UI/UX Tools

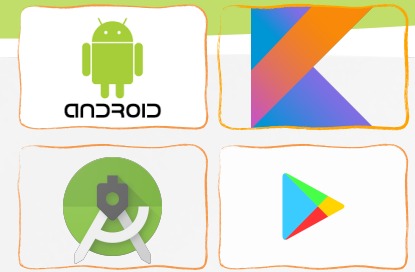
» Sketch: <https://www.sketch.com>

» Adobe XD: <https://www.adobe.com/sea/products/xd.html>

» Figma: <https://www.figma.com>

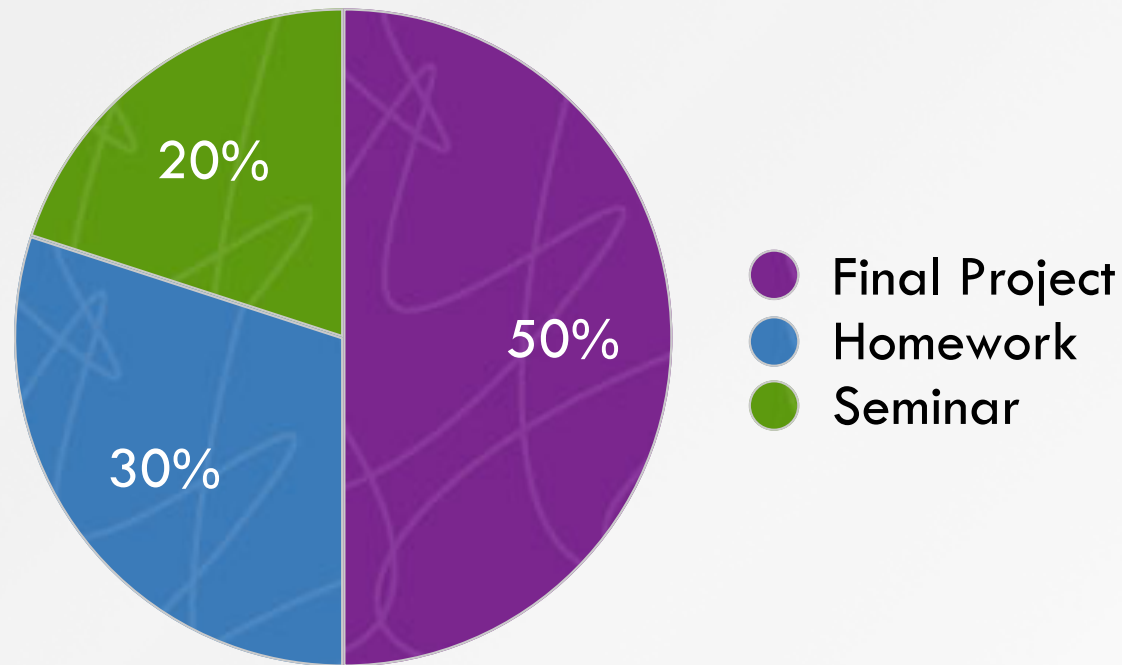
» Invision: <https://www.invisionapp.com>

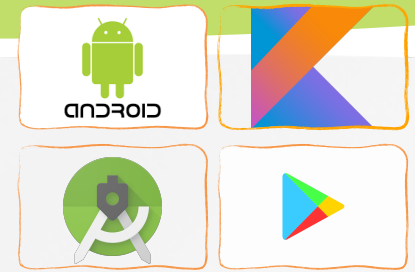




Grades

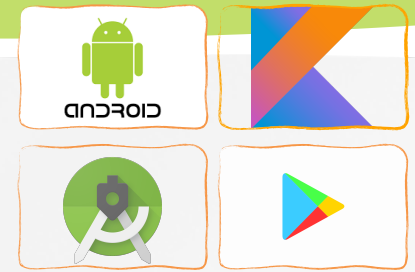
» Course Syllabus...





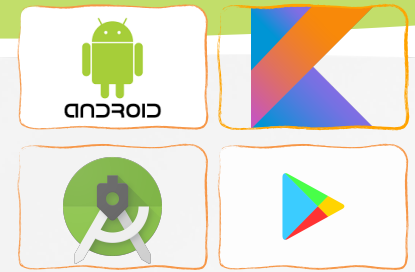
Grades (cont.)

- » gitlab (optional policy)
 - » ~~Assign “nlhdung” as Admin~~
 - » Repo name with format: Android-CLCXX-YY
 - » XX: academic year
 - » YY: group ID
 - » Commit for each new feature
 - » Commit messages must have a prefix:
 - » [+] for adding new feature
 - » [-] for removing a feature
 - » [m] for modifying a feature



Grades (cont.)

- » gitlab (optional policy)
 - » ~~(-0.1)~~ for each empty commit message
 - » Directory Structure
 - » Source
 - » Release
 - » Docs
 - » Videos
 - » Refereces



Policy

- » All students are responsible for reading and following strictly the regulations and policies of the school and university.
- » **For any kind of cheating and plagiarism, students will be graded 0 for the course.** The incident is then submitted to the school and university for further review.
- » Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.
- » **Submission with incorrect name will be graded 0 for that assignment**

Q&A

